

SQL SEQUENCE Operator

Automatic primary key & unique identifier generation, sequence properties, and lifecycle management

1. What is a Sequence?

One-Line Definition: A **SQL Sequence** is a database object used to automatically generate a sequential series of unique numeric values, primarily utilized for auto-generating Primary Keys.

Employee IDs

Transaction Numbers

Application IDs

Registration Numbers

Order Numbers

2. Essential Sequence Properties

Property	SQL Clause	Meaning & Function
Starting Point	START WITH	The initial integer value where sequence generation begins (e.g., 1).
Step Increment	INCREMENT BY	The step size added to the current value for each subsequent request.
Minimum Limit	MINVALUE	The minimum value allowed (useful when decrementing or recycling).
Maximum Limit	MAXVALUE	The upper boundary integer limit allowed for the sequence (e.g., 210).
Recycle Mode	CYCLE	Restarts sequence generation from MINVALUE upon hitting MAXVALUE .
Strict Mode	NOCYCLE	Stops sequence execution and raises a database exception upon hitting MAXVALUE .

3. Creation Syntax & Value Fetching

A. DDL Creation Syntax

```
CREATE SEQUENCE emp_seq START WITH 1 INCREMENT BY 1 MAXVALUE 210 NOCYCLE;
```

4. CYCLE vs NOCYCLE (Crucial Exam Concept)

CYCLE (Looping)

When the sequence reaches **MAXVALUE**, it loops back and restarts generating values from **MINVALUE** (or **START WITH**).

NOCYCLE (Stopping - Default)

When **MAXVALUE** is exceeded, sequence generation terminates immediately and throws a runtime exception error.



Memory Trick: **CYCLE** = Full Dynamic Values (Loop) | **NOCYCLE** = Boundary Hit